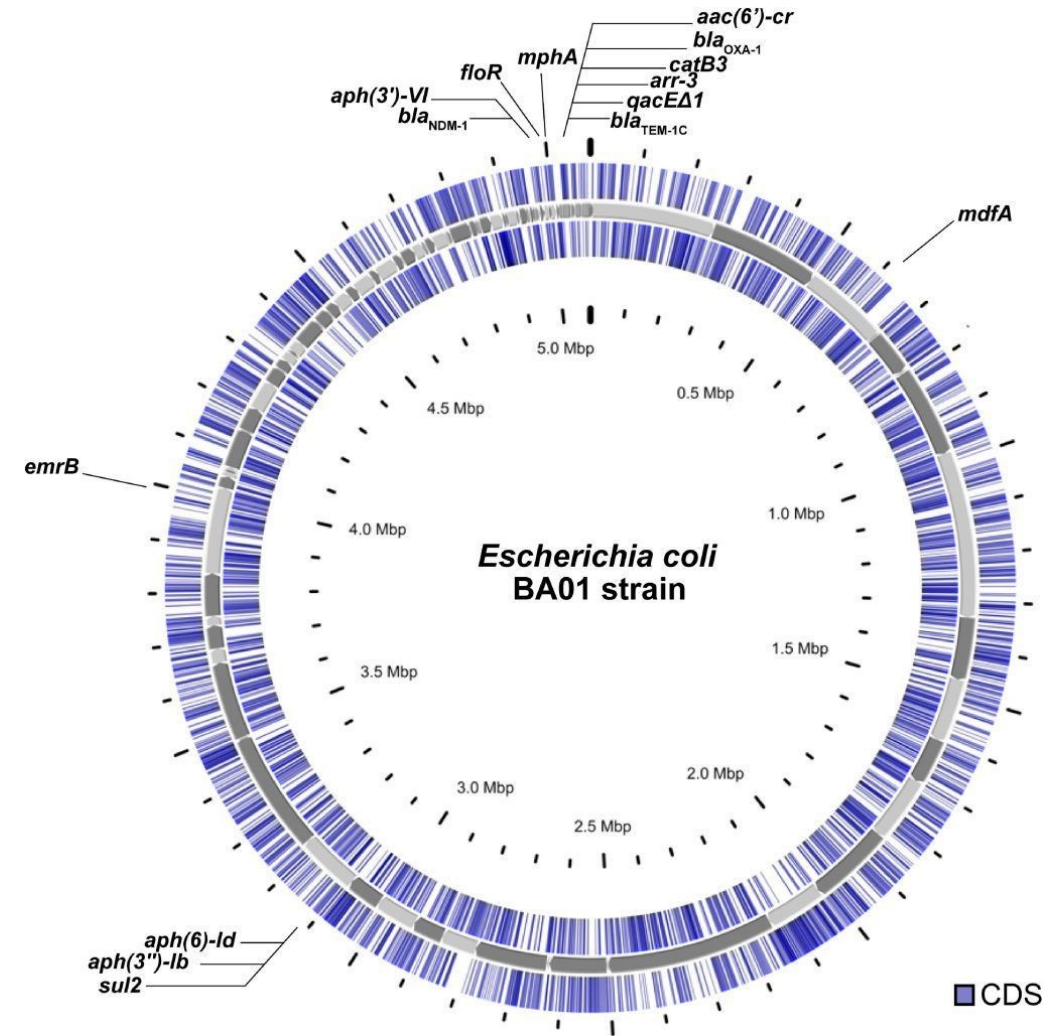
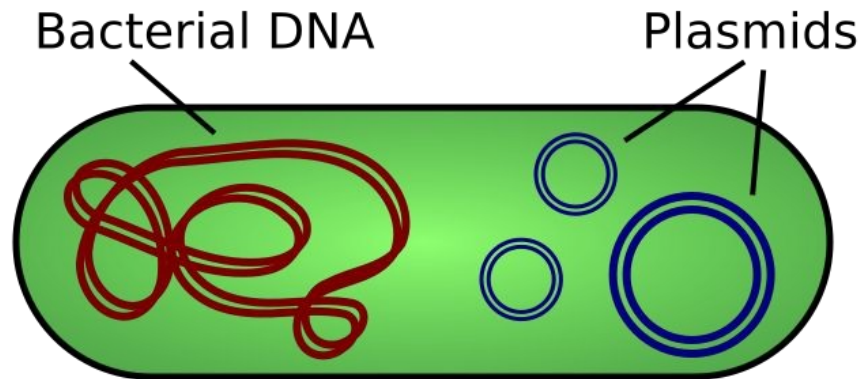


Bacterial Genome Annotation

Shahiid Kiyaga

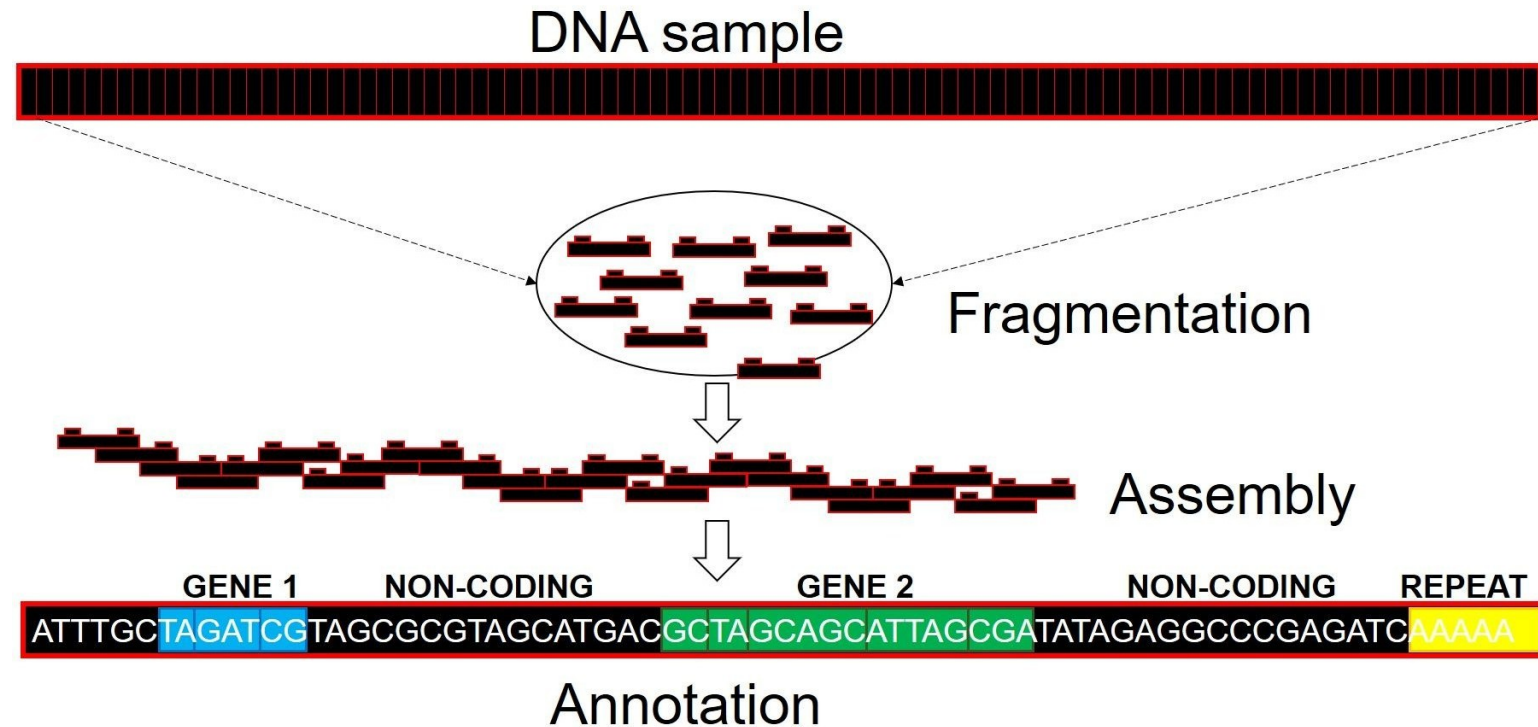
Bacterial Genome

- Circular/linear DNA
- Contains coding & non-coding regions
- Depending on species, 90-95% of the genome may be coding sequence
- Bacteria may also harbor additional extra-chromosomal DNA molecules called plasmids



Assembly vs Annotation

- Assembly: putting reads together → contigs
- Annotation: identifying genes, functions, features



Annotation

- After sequencing and assembly, a genome can be annotated
- Genome annotation describes the structure and function of the components of the genome, by predicting, analyzing, and interpreting them.
- In order to extract their biological significance and understand the biological processes in which they participate.
- Among other things, it identifies the locations of genes and all the coding regions in a genome (structural annotation) and determines what those genes do (functional annotation).

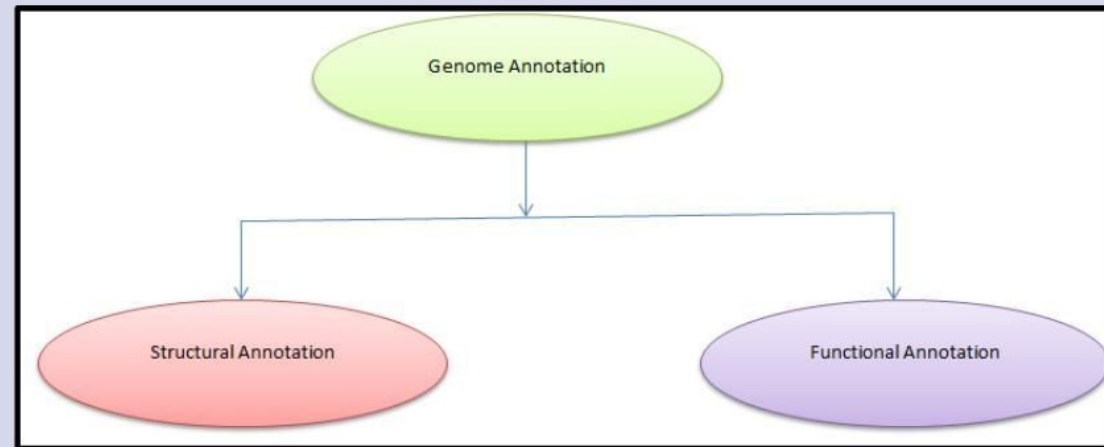
- Why is it important to annotate a genome



Structural vs Functional Annotation

- Structural: find genes, CDS, tRNAs, rRNAs
- Functional: assign roles (enzymes, pathways)

Genome Annotation



gene structure prediction

Identifying elements
(Introns/exons, CDS, stop, start)
in the genome

gene function prediction

Attaching biological information
to these elements- eg: for which
protein exon will code for ⁷

Inputs for Annotation

- Genome assembly (FASTA)
- Reference databases

File formats

- FASTA
- GFF/GTF
- BED

FASTA file format

- Contains nucleotide sequences or amino acid sequences
- A FASTA file can contain one sequence or multiple sequences
- Each FASTA entry begins with a > (greater-than) symbol, followed by a comment on the same line describing the sequence that will follow
- The actual sequence begins on the line after this comment

Examples of fasta file formats

```
>NC_000913.3 Escherichia coli str. K-12 substr. MG1655, complete genome
AGCTTTTCATTCTGACTGCAACGGGCAATATGCTCTGTGTGGATTAAAAAAGAGTGTCTGATAGCAGC
TTCTGAAGTGGTTACCTGCGGTGAGTAAATTTTATTGACTTAGGTCACTAAATACTTTAAACCAA
TATAGGCATAGCGCACAGACAGATAAAAATTACAGAGTACACAACATCCATGAAACGCATTAGCACACC
ATTACCACCACCATCACCATTACCACAGGTAACGGTGGCGGTGACGCGTACAGGAAACACAGAAAAAAG
CCCGCACCTGACAGTGGCGGCTTTTTTTTTCGACCAAGGTAACGAGGTAACACCATGCGAGTGTGAA
GTTCCGGCGGTACATCAGTGGCAAAATGCAGAACGTTTTCTGCGTGTGCGGATATTCTGAAAGCAATGCC
AGGCAGGGGAGGTGGCCACCGTCTCTGCCCCGCCAAAATCACCAACACCTGGTGGCGATGATTG
AAAAAACCATTAGCGGCCAGGATGCTTTACCAATATCAGCGATGCGGCAACGATATTTTGGCCAACTTTT
GACGGGACTCGCCGCCGCCAGCGGGGTTCGCGTGGCGCAATTGAAACTTTCTGTCGATCAGGAATTT
GCCCAATAAAACATGTCCTGCATGGCATTAGTTTGTGGGGCAGTGGCCGATAGCATCAACGCTGCGC
TGATTTGGCGTGGCGAGAAAATGTCGATGCCATTATGGCCGGGATAGAAAGCGCGCGGTCAACAAGT
TACTGTTATCGATCCGGTCGAAAAACTGCTGGCAGTGGGGCATTACCTCGAATCTACCGTCGATATTGCT
GAGTCCACCCCGCGTATTGCGGCAAGCCGATTCCGGCTGATCACATGGTGTGATGGCAGGTTTACCG
CCGGTAATGAAAAGGCGAACTGGTGGTCTTGGACGCAACGGTTCGGACTACTCTGCTGCGGTGCTGGC
TGCGTGTTTACGCCCGGATTGTTGCGAGATTGGACGGGCTGATACCTGCGACCCCGGT
CAGGTGCCCGATTGCGGTTGTTGAAGTCGATGCTTACCAGGAAGCGATGGAGCTTCTACTTCCGGC
CTAAAGTTCTTACCCCGCACCATTAACCCCATCGCCAGTTCCAGATCCCTTGCTGATTAATAATAC
CGGAAATCCTCAAGCACCAGGTACGCTCATTGGTCCAGCCGTGATGAAGACGAATTACCGGTCAAGGGC
ATTTCCAATCTGAATAACATGGCAATGTTACGCGTTTCTGGTCCGGGGATGAAAGGGATGGTCGGCATGG
CGGCGCGCGTCTTTCAGCGATGTCACGCGCCGATTTCCGTTGGTGTGATTACGCAATCATCTTCCGA
ATACAGCATCAGTTTCTGCGTTTCCCAAAGCGACTGTGTCGAGTGAACGGGCAATGCAGGAAGATTTC
TACCTGGAAGTAAAGAGGCTTACTGGAGCCGCTGGCAGTGACGGAACGGCTGGCCATTATCTCGGTGG
TAGGTGATGGTATGCGCACCTTGGTGGGATCTCGCGGAAATCTTTCGGCGACTGGCCGCGGCAATAT
CAACATTGTCGCCATTGCTCAGGGATCTTCTGAACGCTCAATCTCTGCTGGTAAATAACGATGATGCG
ACCATTGGCGTGGCGGTACTCATCAGATGCTTCAATACCGATCAGGTTATCGAAGTGTGTTGATTG
GCGTGGTGGCGTGGCGGTGCGTGTGGAGCAACTGAAGCGTCAGCAAAGCTGGCTGAAGAATAACA
TATCGACTTACGTGTCTGCGGTGTTGCCAACTCGAAGGCTCTGCTACCAATGTACATGGCCTTAATCTG
GAAACTGGCAGGAAGAACTGGCGCAAGCCAAAGAGCCGTTAATCTCGGGCGCTTAATTCGCCTCGTGA
AAGAATATCATCTGCTGAACCCGTCATTGTTGACTGCACCTCCAGCCAGGAGTGGCGGATCAATATGC
CGACTTCTGCGCGGAAGGTTTTCAGGTTGTCACGCGGAAACAAAAGGCCAACACCTCGTGCATGGATTAC
```

```
>AJJDOH_00005 adenine deaminase
TTGTTAGCCGTTTCCCGTGGCGACGCTGTTGCCGATTATATTATTGATAATGTCTCTATTCTCGACCTGATCAATGGCGGAGAAATTTCCGGCCCAATTGTGATTAAAGGACGTTACATTGC
>AJJDOH_00010 hexose phosphate transporter UhpT
ATGCTGGCTTTCTTAAACCAGGTTTCGAAGCCGACCTGGACCTTCCGCTCGAAGTGGCGGCAAAATGTGGTTCAAACCGTTTCATGCAATCCTACCTGGTGGTCTTTATCGGCTACCTGAC
>AJJDOH_00015 MFS transporter family glucose-6-phosphate receptor UhpC
ATGTTGCCGTTTCTGAAAGCGCTGCGGATGCGCCATTAATGACTGATAAATATGAAATTGATGCCCGCTATCGCTACTGGCGTGGCGATATTCTGCTGACCATCTGGCTGGGTTACGCGCT
>AJJDOH_00020 signal transduction histidine-protein kinase/phosphatase UhpB
ATGAAGACGTTGTTCTCCGCTTAATTACCGTTATTGCTGCTTTTATCTTCTCTGCGCATGGTTTGCCTGTGGAGTATCAGCCTGCACCTGGTTGAGCGCCCTGATATGGCGGTGCT
>AJJDOH_00025 DNA-binding response regulator UhpA
ATGATCACCGTTGCCCTTATAGACGATCACCTCATCGTCCGCTCCGGCTTGGCGAGTGTGGGGCTGGAACCTGATTGACAGTAGTTGCCAGTTTGGTTGGGGCGCGAGGCGCTGGC
>AJJDOH_00030 hypothetical protein
GTGGCCTTTTGGCCGCGCGCTTTAACGTTGAAGGCATTCTTTGTNNNNNNNNNATTACAGACAGCGACAAAAGCCATATCTGGCTACTGGTCAATGACGACCAGCGTCTGGAGCAGAT
>AJJDOH_00035 Acetolactate synthase
ATGGCAAGTTCGGGCACAACATCGACGCGTAAGCGCTTACCAGCGCAGAAATTTATCGTTCAATTCCTGGAACAGCAGGGCATTAAAGATTGTACGGGCATTCCGGGCGGTTCTATCCTGCC
>AJJDOH_00040 iStr Hfq binding RNA
CGTCAAAATTCGTGCCGAAATGGCGGTTCTGCGCGGAACACGTATACTTTCAGTGTTGACATAATACAGTGTGCTTTCGGGTTACCAGCCGAGGCGGCTGATGAAACCTCGCTCCGGCGG
>AJJDOH_00045 Small toxic protein TisB
ATGAGCCTGGTGGATATCGCCATTCTATCTCAAACCTATTGTTGCAGCACTGCAACTGCTTGATGCTGTTCTGAAATACCTGAAGTAA
>AJJDOH_00050 Protein YsdE
ATGAATGTTAGCCAGATATACGCCAGGAATGGGGAATTGTTAGCGGCAGAAATATGTAACAAAAGCGGCAATAA
>AJJDOH_00055 Multidrug resistance protein D
ATGAAAAGGCAAAGAAACGTCATTTGTTATTGATGTTGGTATTACTCGTGGCGTGGTTCAGATGGCGCAACCATTTATATCCAGCTATTGCCGATATGGCGCGGATCTCAACGCTCGC
>AJJDOH_00060 Sulfatase maturation enzyme AslB, radical SAM superfamily
ATGACAGGAAGTCAGGTCAATGATGCTGAAGAGGACAGGCATAAGCTGGTTGATAGATATAAGGATGCTCTGCAACCTGCCGATTTTATCACAATTTCAAACAACGGGGCATCCGCTCCGT
```

- Single sequence

- Multiple sequences

FASTA FILENAME EXTENSIONS

- FASTA files usually end with the extension .fasta. This extension is arbitrary, as the content of the file determines its format, not its extension.
- More descriptive filename extensions can be used instead of .fasta, which are useful as they describe the type of sequence(s) in the file at a glance.
- Here are some examples...
- .fna can be used for FASTA nucleic acids
- .faa can be used for FASTA amino acids
- .frn can be used for FASTA non-coding RNA

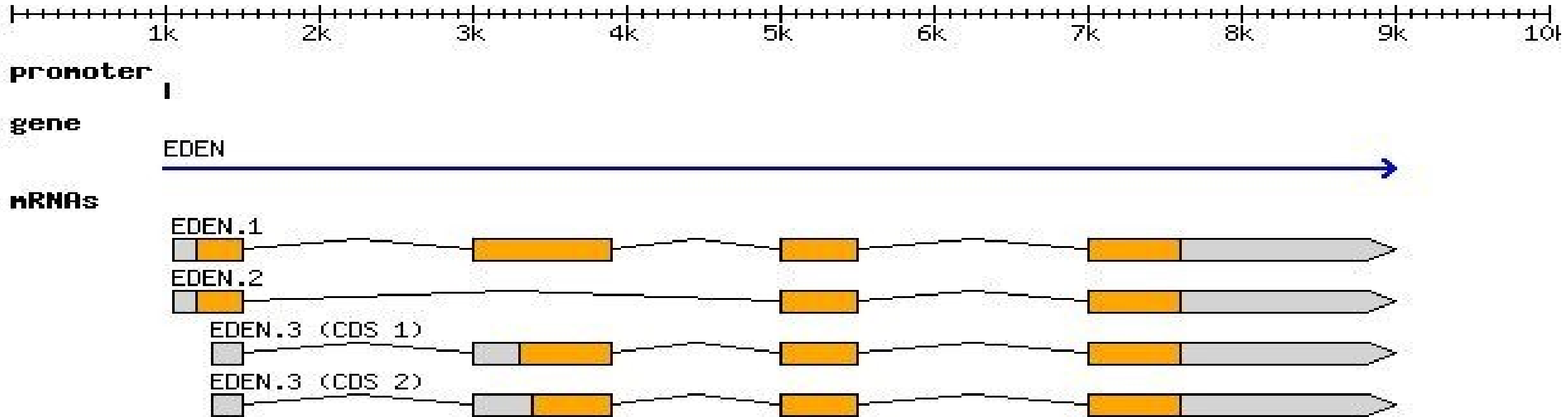
GFF/GTF File Format

- The GTF (Gene Transfer Format) and GFF (General Feature Format) file formats are commonly used in bioinformatics and genomics to represent and store information about genomic features and annotations, such as genes, transcripts, exons, and other genomic elements.
- These formats are plain-text and tab-delimited, making them human-readable and easy to work with programmatically
- The GFF (General Feature Format) format consists of one line per feature, each containing 9 columns of data, plus optional track definition lines

GFF/GTF fields

- All but the final field in each feature line must contain a value; "empty" columns should be denoted with a '.'
1. Sequence ID
 2. Source - Describes the algorithm or the procedure that generated this feature. Typically Genescane or Genebank, respectively.
 3. Feature Type - Describes what the feature is (mRNA, domain, exon, etc.).
 4. Feature Start
 5. Feature End
 6. Score - Typically E-values for sequence similarity and P-values for predictions.
 7. Strand
 8. Phase - Indicates where the feature begins with reference to the reading frame. The phase is one of the integers 0, 1, or 2, indicating the number of bases that should be removed from the beginning of this feature to reach the first base of the next codon.
 9. Attributes - A semicolon-separated list of tag-value pairs, providing additional information about each feature. Some of these tags are predefined, e.g. ID, Name, Alias, Parent

Example



GFF/GTF fields

- Lines beginning with '##' are metadata and provide meta-information about the document as a whole. Blank lines should be ignored by parsers and lines beginning with a single '#' are used for human-readable comments and can be ignored by parsers.
- Line 0 gives the GFF version. Line 1 indicates the boundaries of the region being annotated (a 1,497,228 bp region named "ctg123").
- Line 2 defines the boundaries of the gene. Column 9 of this line assigns the gene an ID of gene00001, and a human-readable name of EDEN. Because the gene is not part of a larger feature, it has no Parent.
- Line 3 annotates the transcriptional factor binding site. Since it is logically part of the gene, its Parent attribute is gene00001.

GFF/GTF fields

- Lines 4-6 define this gene's three spliced transcripts, one line for the full extent of each of the mRNAs. These features are necessary to act as parents for the four CDSs which derive from them, as well as the structural parents of the five exons in the alternative splicing set.
- Lines 7-11 identify the five exons. The Parent attributes indicate which mRNAs the exons belong to. Notice that several of the exons share the same parents, using the comma symbol to indicate multiple parentage.
- Lines 12-24 denote this gene's four CDSs. Each CDS belongs to one of the mRNAs. cds00003 and cds00004, which correspond to alternative start codons, belong to the same mRNA.

GFF/GTF fields

- Note that several of the features, including the gene, its mRNAs and the CDSs, all have Name attributes.
- This attribute assigns those features a public name, but is not mandatory.
- The ID attributes are only mandatory for those features that have children (the gene and mRNAs), or for those that span multiple lines.
- The IDs are not required to have meaning outside the file in which they reside.
- Hence, a slightly simplified version of this file would look like this

```
##gff-version 3.1.26
##sequence-region ctg123 1 1497228
ctg123 . gene 1000 9000 . + . ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000 1012 . + . Parent=gene00001
ctg123 . mRNA 1050 9000 . + . ID=mRNA00001;Parent=gene00001
ctg123 . mRNA 1050 9000 . + . ID=mRNA00002;Parent=gene00001
ctg123 . mRNA 1300 9000 . + . ID=mRNA00003;Parent=gene00001
ctg123 . exon 1300 1500 . + . Parent=mRNA00003
ctg123 . exon 1050 1500 . + . Parent=mRNA00001,mRNA00002
ctg123 . exon 3000 3902 . + . Parent=mRNA00001,mRNA00003
ctg123 . exon 5000 5500 . + . Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon 7000 9000 . + . Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS 1201 1500 . + 0 ID=cds00001;Parent=mRNA00001
ctg123 . CDS 3000 3902 . + 0 ID=cds00001;Parent=mRNA00001
ctg123 . CDS 5000 5500 . + 0 ID=cds00001;Parent=mRNA00001
ctg123 . CDS 7000 7600 . + 0 ID=cds00001;Parent=mRNA00001
ctg123 . CDS 1201 1500 . + 0 ID=cds00002;Parent=mRNA00002
ctg123 . CDS 5000 5500 . + 0 ID=cds00002;Parent=mRNA00002
ctg123 . CDS 7000 7600 . + 0 ID=cds00002;Parent=mRNA00002
ctg123 . CDS 3301 3902 . + 0 ID=cds00003;Parent=mRNA00003
ctg123 . CDS 5000 5500 . + 1 ID=cds00003;Parent=mRNA00003
ctg123 . CDS 7000 7600 . + 1 ID=cds00003;Parent=mRNA00003
ctg123 . CDS 3391 3902 . + 0 ID=cds00004;Parent=mRNA00003
ctg123 . CDS 5000 5500 . + 1 ID=cds00004;Parent=mRNA00003
ctg123 . CDS 7000 7600 . + 1 ID=cds00004;Parent=mRNA00003
```

BED file format

- The BED format (Browser Extensible Data) is a simple and compact tab-delimited format that is commonly used to represent genomic intervals, such as regions, or features of interest.
- It is often used in genome browser applications and for defining regions in genomic data.
- A standard BED file consists of three to twelve columns, with the first three columns being required:

1. **Chromosome Name** - Name of the chromosome or scaffold.
2. **Chromosome Start** - Start position of the feature in standard chromosome base is 0).
3. **Chromosome End** - End position of the feature in standard chromosome

```
chr1 213941196 213942363
chr1 213942363 213943530
chr1 213943530 213944697
chr2 158364697 158365864
chr2 158365864 158367031
chr3 127477031 127478198
chr3 127478198 127479365
chr3 127479365 127480532
chr3 127480532 127481699
```

BED file format

Nine additional fields are optional. Note that columns cannot be empty – lower-numbered fields must always be populated if higher-numbered ones are used.

1. **Name** - Label to be displayed under the feature.
2. **Score** - A score between 0 and 1000.
3. **Strand** - defined as + \(\text{forward}\) or – \(\text{reverse}\).
4. **thickStart** - coordinate at which to start drawing the feature as a solid rectangle
5. **thickEnd** - coordinate at which to stop drawing the feature as a solid rectangle
6. **itemRgb** - an RGB colour value \(\text{e.g. } 0,0,255\). Only used if there is a track line with the value of itemRgb set to “on” \(\text{case-insensitive}\).
7. **blockCount** - the number of sub-elements \(\text{e.g. exons}\) within the feature
8. **blockSizes** - the size of these sub-elements
9. **blockStarts** - the start coordinate of each sub-element

Annotation tools

- Bakta
- RAST
- Proka
- etc

What next

- Comparative genomics: Guided by question at hand